

The Ethical Hacker's Command Handbook

Chapter 2 – Part 4: Nmap Scripting Engine (NSE)

Automate enumeration with Nmap's built-in Lua scripts.

Learning Objectives

Understand what NSE is, use default and custom scripts, search for scripts, learn common categories, and know when each script is appropriate during authorized security assessments.

What is NSE?

NSE (Nmap Scripting Engine) extends Nmap using Lua scripts to automate information gathering and selected security checks. Instead of only showing open ports, scripts can retrieve banners, web page titles, SMB details, SSL information, and more.

Why Use NSE?

After discovering open ports and services, NSE helps collect additional information that can guide the next stage of enumeration or validation.

nmap -sC

Purpose: Default scripts

What it does: Runs Nmap's default script set.

When to use: After identifying open services.

Avoid when: When only a very quick scan is needed.

Short Note: Next: Review results and research.

nmap --script http-title

Purpose: HTTP title

What it does: Retrieves the title of a web page.

When to use: When HTTP/HTTPS is open.

Avoid when: No web service detected.

Short Note: Useful for identifying applications.

nmap --script ftp-anon

Purpose: FTP anonymous

What it does: Checks for anonymous FTP access.

When to use: When FTP (21) is open.

Avoid when: FTP not present.

Short Note: Verify access is authorized.

nmap --script smb-os-discovery

Purpose: SMB OS discovery

What it does: Collects SMB OS and host information.

When to use: When SMB (445) is open.

Avoid when: SMB unavailable.

Short Note: Useful in Windows environments.

nmap --script ssh-hostkey

Purpose: SSH host key

What it does: Retrieves SSH public host keys.

When to use: When SSH (22) is open.

Avoid when: SSH unavailable.

Short Note: Useful for fingerprinting.

nmap --script vuln

Purpose: Vulnerability category

What it does: Runs scripts in the vuln category.

When to use: After identifying services.

Avoid when: Outside the authorized scope.

Short Note: Validate findings before conclusions.

nmap --script smb-vuln-ms17-010 -p445

Purpose: MS17-010 check

What it does: Checks for indicators of the MS17-010 SMB vulnerability.

When to use: When Windows SMB is exposed.

Avoid when: Port 445 closed.

Short Note: Research results before next steps.

sudo nmap --script-updatedb

Purpose: Update script database

What it does: Rebuilds the local NSE script database.

When to use: After adding or modifying scripts.

Avoid when: Routine scans.

Short Note: Keeps local script index current.

Finding Scripts

List scripts: `ls /usr/share/nmap/scripts`

Search: `ls /usr/share/nmap/scripts | grep smb`

NSE Categories

Category	Purpose
default	Common safe scripts
safe	Low-impact checks
discovery	Gather information
auth	Authentication-related checks
brute	Credential guessing scripts
vuln	Vulnerability checks
malware	Malware detection
exploit	Exploit-related scripts

Workflow

Discover host → Scan ports → Detect versions → Run -sC → Run targeted scripts → Research → Document findings

Revision Notes

- -sC = Default scripts
- --script = Specific script
- http-title = Page title
- ftp-anon = Anonymous FTP check
- smb-os-discovery = SMB information
- ssh-hostkey = SSH host keys
- --script vuln = Vulnerability scripts
- --script-updatedb = Refresh script database

Practice Questions

What is the difference between -sC and --script?

When would you use http-title?

Why should services be identified before running targeted NSE scripts?

What does smb-os-discovery provide?